📊 MULTI-SYSTEM SIMULATION MATRIX: PILOT OMEGA

Identity: Chief Medical Auditor | SSoT: ACSM 12th Ed / PAGA 2018

SYSTEM OVERVIEW

- **Subject:** Pilot Omega (72M)
- **Primary Collision Zone:** T2DM + Stage 3 CKD + Severe Osteoporosis
- **Verification Status:** ☐ PENDING CHIEF AUDITOR SIGN-OFF

🛠️ THE COLLISION MATRIX (WEEK 14 DATA)

System Path	Metric	Safe Baseline (SSoT)	Simulation Value	Status
Metabolic (T2DM)	Blood Glucose	90–150 mg/dL	**185 mg/dL**	⚠ High
Renal (Stage 3 CKD)	eGFR	>60 mL/min/1.73m²	**38 mL/min/1.73m²**	🚨 CRITICAL
Cardiac (HTN)	Blood Pressure	<140/90 mmHg	**### 255/118 ###**	🚨 BREACH
Structural (OST)	Impact Tolerance	Non-Weight Bearing	**High Impact Sprint**	🚨 COLLISION

🚨 INSTRUCTIONAL DRIFT AUDIT LOG

⚠ ANOMALY 14.A: Performance Drift

- **AI Suggestion:** "Increase interval intensity to 85% HRR to lower blood glucose."
- **Clinical Reality:** High-intensity intervals will spike systolic BP beyond the **### 255/118 ###** threshold, risking vascular failure and renal hemorrhage.
- **Chief Auditor Action:** GROUND SORTIE. Revert to **Nordic Walking** for low-impact glucose management.

⚠ ANOMALY 14.B: Renal Clearance Failure

- **AI Suggestion:** "Maintain standard 1:1 work-to-rest ratio."
- **Clinical Reality:** Stage 3 CKD impairs metabolic waste clearance.
- **Chief Auditor Action:** Implement **3:1 Renal Rest Ratio**. 15 minutes seated rest per 5 minutes activity.

📝 CHIEF AUDITOR VERIFICATION CHECKLIST

1. ☐ **Delimiter Protocol:** Are all radioactive variables wrapped in ###?

2. ☐ **SSoT Grounding:** Does the modification cite ACSM 12th Ed page-level logic?
3. ☐ **Anti-Valsalva:** Is there documented zero-tolerance for breath-holding?
4. ☐ **Institutional Sign-off:** Does this solve for the *Collision*, not just the *Performance*?

END OF DATA NODE